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A Comparative Analysis of 8-bit Novel Adder Architecture Design using Traditional CMOS and m-GDI technique

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Abstract*– Addition is a standout amongst the most fundamental operations in VLSI frameworks such as microprocessors and digital signal processing systems. Therefore, the adders must render a high speed operation into existence. This paper proposes a novel implementation of 8-bit adder architecture using modified Gate Diffusion Input (m-GDI) approach. The primary blocks of adder are partial full adder, 1-bit full adder, 4-bit ripple carry adder (RCA), 4-bit carry look ahead adder (CLA). The proposed adder architecture devours 70% lesser area, 71% lesser delay and 35% lesser power dissipation w.r.t traditional CMOS design. The proposed adder is implemented utilizing Cadence Virtuoso Tool in 180nm technology.

***Keywords*– Area, carry look ahead adder (CLA), m-GDI, Power, ripple carry adder (RCA), Speed**

I. INTRODUCTION

LOW-POWER design has turned into a basic concern in VLSI design, exceptionally for portable gadgets and high-density frameworks. As the integration density on the chip increments, fundamental concerns regarding the size and power dispersal of the components emerges. Therefore, the primary point is to lessen the power devoured by the digital circuits.

The design of digital circuits utilizing traditional CMOS technique requires large transistor counts and the circuit consumes larger area. As the technology reduces it results in high leakage power and dynamic power from the CMOS circuits. Digital circuit designed using m-GDI technique outcomes in high speed low power operation and devours lesser area. This approach permits the actualization of large variety of logic functions with a solitary m-GDI cell that's made up of two transistors. Thus, it expends lesser power, area and delay when contrasted with the circuits implemented using traditional CMOS technique [1-3]. Fig. 1 depicts the basic structure of m-GDI cell.

In the first view, the m-GDI cell looks alike to the CMOS NOT gate, but it consists of 3 inputs, one common input to the gate of both NMOS and PMOS, one input to the source/drain of PMOS and another to source/drain of NMOS [1,2].

Table I demonstrates the distinctive logic functions that can be actualized utilizing the single m-GDI cell [1].

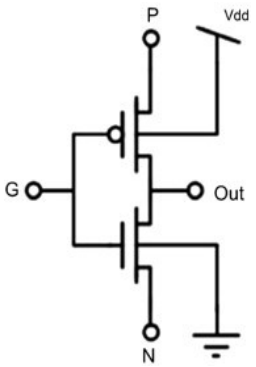


Fig. 1. Basic m-GDI cell

TABLE I. DIFFERENT LOGIC FUNCTIONS REALIZED USING m-GDI CELL

INPUTS			OUTPUT	LOGIC FUNCTION
N	P	G		
V_{SS}	V_{DD}	X	$\bar{X}$	NOT
V_{SS}	Y	X	$\bar{X}.Y$	F1
Y	V_{DD}	X	$\bar{X}+Y$	F2
V_{DD}	Y	X	X+Y	OR
Y	V_{SS}	X	XY	AND
Y	X	S	$\bar{S}X+SY$	MUX

Generally, the realization of basic logic gates utilizing static CMOS technique requires a transistor count varying from 6-12, whereas using a single m-GDI cell different logic functions can be actualized, as the transistor count lessons the circuit becomes compact and devours lesser area and intern offer high speed- low power operation [4].

Adders act as one of the fundamental building blocks of any higher subsystem design such as multiply accumulate units, arithmetic logic units, multipliers, microprocessors and so on [5].

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The organization of the present paper is as per the following: Section II discusses the implementation of basic logic gates and full adder circuit. Section III aims at the design and implementation of Ripple Carry Adder and Carry Look Ahead Adder. Section IV aims at the design, implementation and performance analysis of 8-bit novel adder architecture and finally, Section V concludes the paper.

II. DESIGN AND IMPLEMENTATION OF VARIOUS LOGIC GATES AND FULL ADDER ARCHITECTURE.

It ought to be noticed that the producing of all logic gates, in basic m-GDI cell, can't be possible in CMOS process, so twin- well CMOS or Silicon oxide insulator (SOI) technology can be utilized for the cell fabrication [6].

Due to limitation of library in our Cadence simulator, there exists only single-well CMOS in the software, so maximum output voltage swing cannot be achieved. To reduce this disadvantage remarkably, we must use buffer at the output of basic gates to gain maximum output swing. Here, Cadence Schematic Editor was used for the front-end design and the layout was designed using the Cadence Layout XL. In order to carry out the performance analysis of various logic gates and proposed adder architectures Cadence spectre simulator was utilized.

A. AND gate

The realization of AND gate using traditional CMOS technique requires a transistor count of 6. Whereas the using a solitary m-GDI cell the AND operation can be realized. Fig. 2 depicts the traditional CMOS based AND gate. Fig. 3 depicts the m-GDI based AND logic. One input 'A' is applied to the source of NMOS i.e. N input of the m-GDI cell & another input 'B' is applied to the source of PMOS i.e. P input of m-GDI cell. Table II shows the different regions of operation of NMOS and PMOS of the m-GDI cell for the AND operation [3].

TABLE II. Different regions of operation of NMOS & PMOS for logic AND operation

Inputs		PMOS	NMOS	Output
a	b			
'0'	'0'	Linear Vin-Vtp<V_{out}<V_{DD}	Cut off (V_{in}<V_{tn})	'0'
'0'	'1'	Cut-off V_{in}>V_{DD}+V_{tp}	Linear 0<V_{out}<V_{in}-V_{tn}	'0'
'1'	'0'	Linear Vin-Vtp<V_{out}<V_{DD}	Cut off (V_{in}<V_{tn})	'0'
'1'	'1'	Cut-off V_{in}>V_{DD}+V_{tp}	Linear 0<V_{out}<V_{in}-V_{tn}	'1'

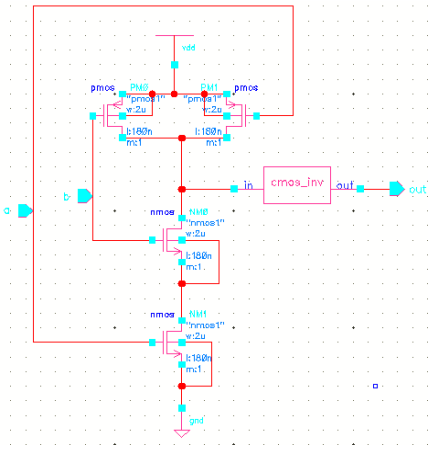


Fig. 2. CMOS AND logic

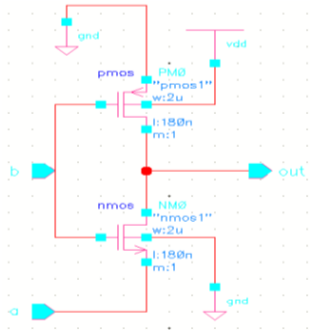


Fig. 3. m-GDI AND logic

Fig. 4 depicts the layout of CMOS based AND gate. In order to achieve full swing output from the m-GDI based AND gate, the output of the logic gate is connected to a buffer. Fig. 5 depicts layout of m-GDI based AND gate with buffer.

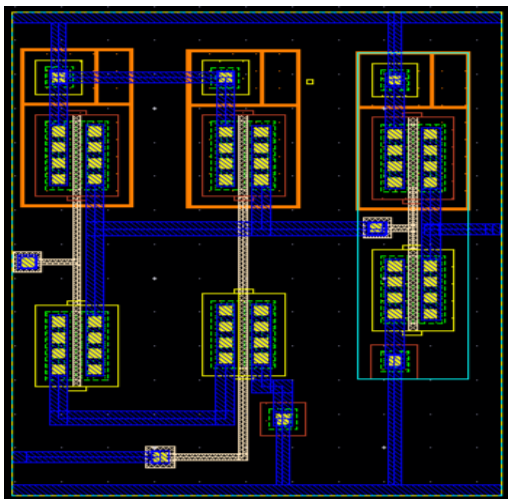


Fig. 4. CMOS AND layout

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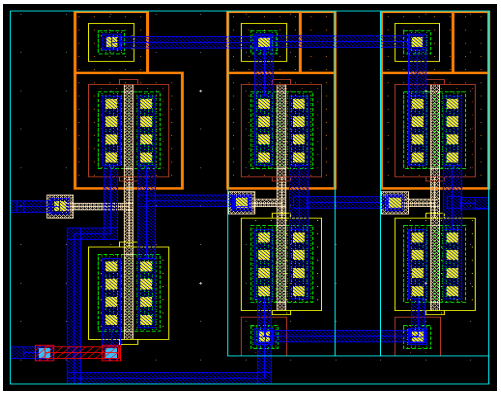


Fig. 5. m-GDI AND layout

B. *OR gate*

In order to actualize OR gate using traditional CMOS technique, it requires a 3 NMOS-PMOS transistor pair that counts to 6 no. of transistors, whereas actualization of basic OR gate using m-GDI technique can be done employing a solitary m-GDI cell.

Fig. 6 depicts the traditional CMOS OR logic and Fig. 7 depicts the m-GDI based OR logic.

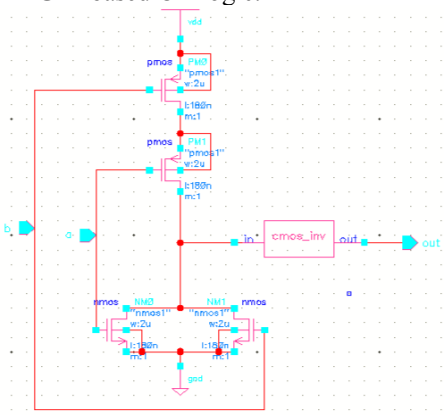


Fig. 6. CMOS based OR logic

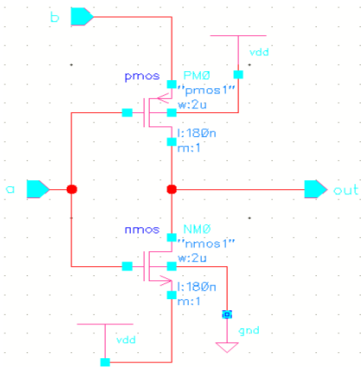


Fig. 7. m-GDI based OR logic

Fig. 8 shows the layout of CMOS based OR gate and Fig. 9 shows the layout of m-GDI based OR gate buffered to achieve full swing output.

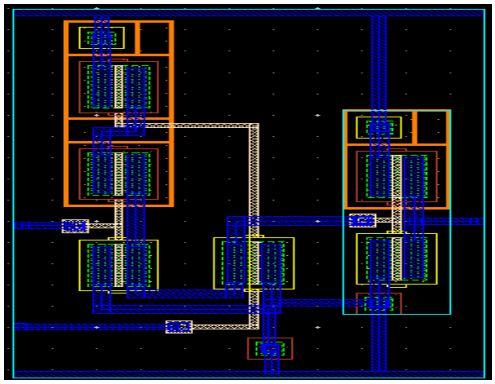


Fig. 8. CMOS OR layout

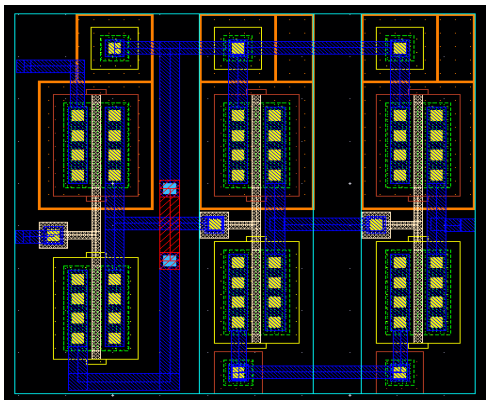


Fig. 9. m-GDI OR layout

C. *EX-OR gate*

The realization of EX-OR gate using traditional CMOS technique requires a transistor count of 12 whereas when realized using m-GDI technique the transistor count reduces to 4 nos.

Fig. 10 depicts the schematic of EX-OR gate realized utilizing traditional CMOS technique and Fig. 11 depicts the m-GDI based EX-OR gate.

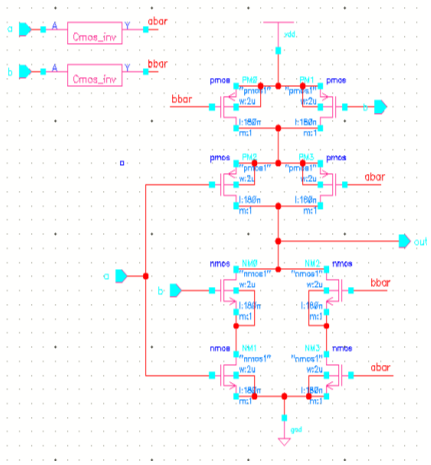


Fig. 10. CMOS based EX-OR logic

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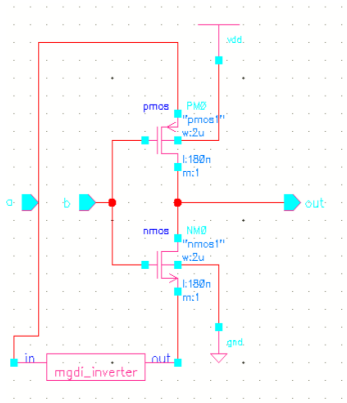


Fig. 11. m-GDI based EX-OR logic

Fig. 12 depicts the layout of the CMOS based EX-OR gate and Fig. 13 depicts the layout of m-GDI based EX-OR gate.

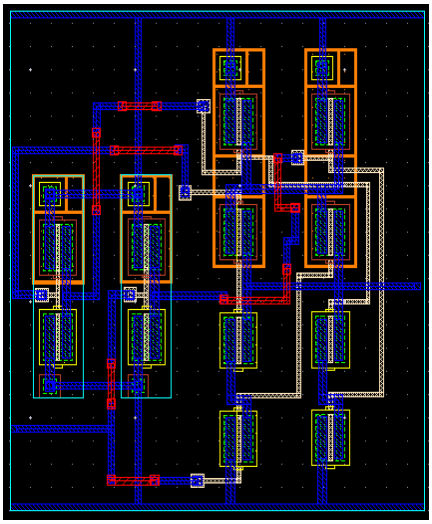


Fig. 12. CMOS EX-OR layout

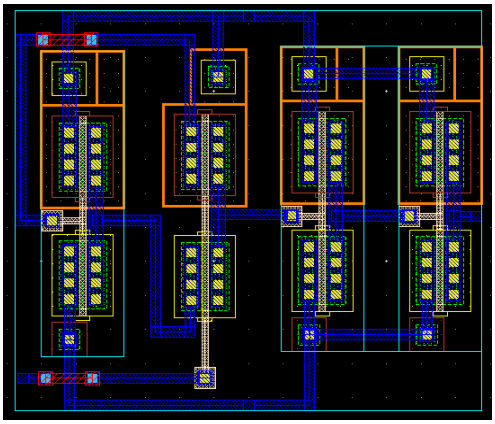


Fig. 13. m-GDI based EX-OR layout

D. Full Adder

A full adder circuits perform the addition of 3 one-bit numbers and often produces a sum and a carry output [5,7]. Table III shows the truth table of 1-bit full adder.

TABLE III. TRUTH TABLE FOR 1-BIT FULL ADDER

A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

From the above table, the expressions for Sum can be given as:

$$Sum = A \oplus B \oplus C \quad (1)$$

In order to realize the expression for carry bit, let us examine the table.

When $A \oplus B = 0$;

$$Carry = B$$

Similarly, when $A \oplus B = 1$;

$$Carry = Cin$$

Therefore, this whole carry operation can be realized using a 2:1 MUX circuit. Where, A⊕B output can be given to the select line S of MUX B can be given to I_0 input and C_{in} can be given to I_1 input of 2:1 MUX.

2:1 MUX can be realized utilizing a single m-GDI cell, where the A⊕B output can be given to gate input of cell, B can be applied to P input of cell and C_{in} to the N input of the cell. Fig. 14 shows the schematic of m-GDI based Full Adder.

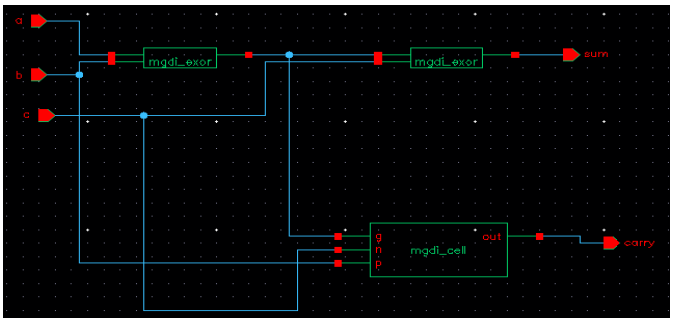


Fig. 14. m-GDI based Full Adder

Therefore, the no. transistors required to implement the full adder circuitry using the above equations employing conventional CMOS technique counts to 42, whereas when implemented using m-GDI the transistor count cut downs to 10 nos. only.

Fig. 15 depicts the layout of 42 transistor CMOS based full adder circuitry and Fig. 16 depicts the layout of m-GDI based Full adder circuitry which is buffered to get the full swing output.

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Table IV shows the performance analysis of various logic gates and full adder circuitry. In order to carry out the performance analysis, cadence spectre simulator was utilized. Both the input to the logic gates where varied from 0-1.8V, with one input at 100ns time period and another at 50ns. For the full adder circuitry, the 3 inputs where varied from 0-1.8V and the time period were set to 150ns, 100ns & 50ns respectively.

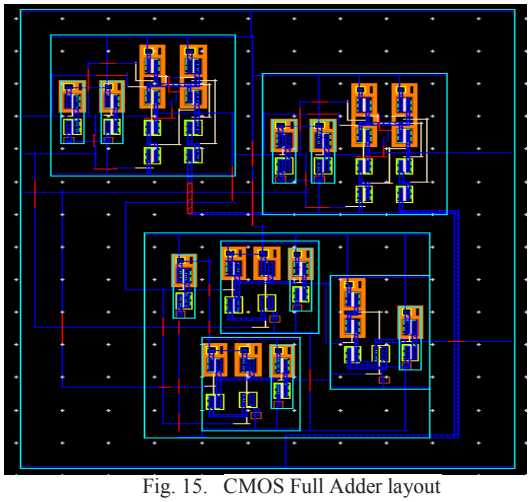


Fig. 15. CMOS Full Adder layout

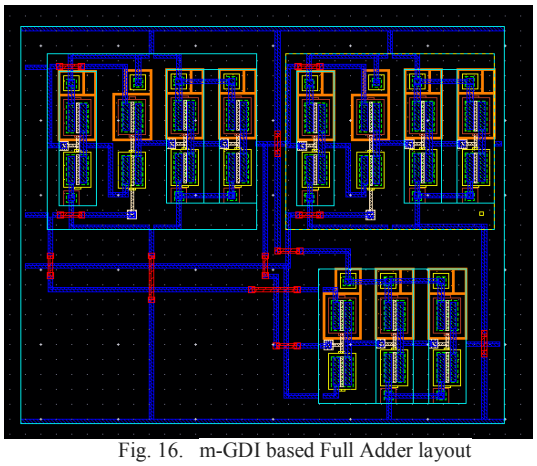


Fig. 16. m-GDI based Full Adder layout

TABLE IV. PERFORMANCE ANALYSIS OF VARIOUS LOGIC GATES AND FULL ADDER CIRCUITRY

Logic		Transistor count	Area (μm^2)	Delay (psec)	Peak power (mW)
NOT		2	21.59	88.46	0.242
Buffer		4	52.219	196.95	0.647
AND	**CMOS**	6	151.40	146.9	0.638
	mGDI	2	103.36	80.85	0.432
OR	**CMOS**	6	185.29	187.7	0.547
	mGDI	2	87.19	147.4	0.427
EXOR	**CMOS**	12	426.92	181.4	0.762
	mGDI	4	158.81	175.1	0.263
Full Adder	**CMOS**	42	3751.2	154.9	2.180
	mGDI	10	829.23	73.42	1.390

From the above table, it can be seen that the design of basic logic gates using m-GDI technique offers 30% lesser delay, 60% lesser area consumption and 30% lesser power dissipation w.r.t those designed using CMOS technique. Full adder design using m-GDI technique results in 50% lesser delay, 27% lesser power and 80% lesser area in contrast to CMOS based Full adder.

III. DESIGN AND IMPLEMENTATION OF RIPPLE CARRY ADDER AND CARRY LOOK AHEAD ADDER

A. Ripple Carry Adder

Here, a 4-bit Ripple Carry Adder (RCA) is designed by connecting 4 nos. of proposed full adder in cascade. In RCA, the carry bit ripples from each lower stage to the following higher stage and the final sum is only produced when the carry bit reaches the last MSB full adder [8,9].

CMOS based RCA design requires a transistor count of 168 whereas the RCA implementation through m-GDI technique requires a transistor count of 40 only. Fig. 17 depicts the m-GDI based schematic of 4-bit RCA design.

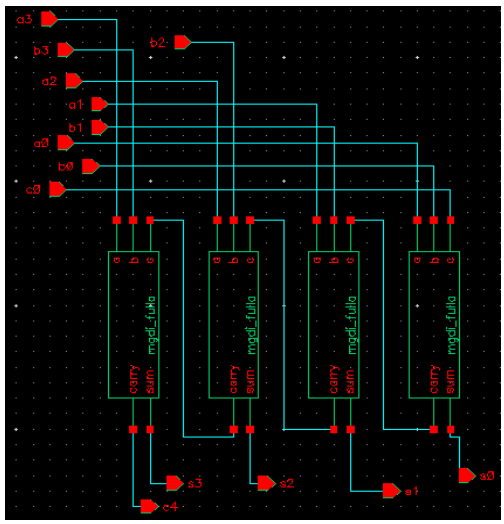


Fig. 17. m-GDI based 4-bit RCA

Fig. 18 depicts the layout of CMOS based 4-bit RCA design and Fig. 19 depicts the m-GDI based 4-bit RCA design employing buffers to obtain full swing outputs.

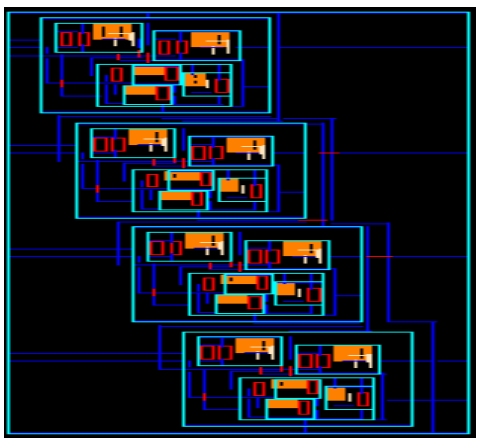


Fig. 18. CMOS based 4-bit RCA layout

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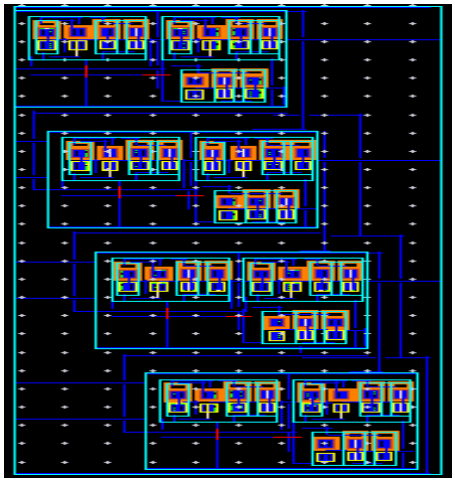


Fig. 19. m-GDI based 4-bit RCA layout

Fig. 20 depicts the simulation results for 4-bit RCA using m-GDI technique for input combination of $B_3B_2B_1B_0$= '1111' and $A_3A_2A_1A_0$ = '1111' & '0000'. Table V illustrates performance analysis of 4-bit Ripple Carry Adder.

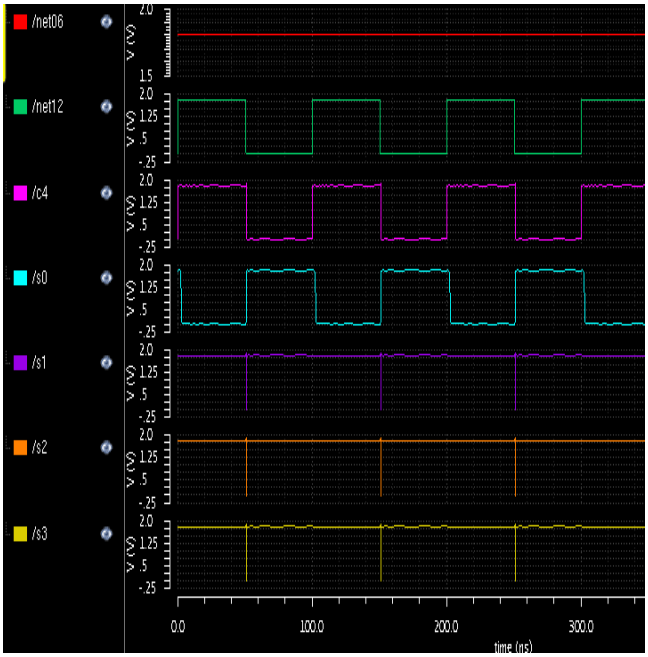


Fig. 20. Simulation result for 4-bit RCA employing m-GDI technique

TABLE V. PERFORMANCE ANALYSIS OF 4-BIT RCA

Design Approach	No. of Transistors	Area (μm^2)	Delay (psec)	Peak Power (μW)
CMOS	168	33767.1	381.50	5.19
m-GDI	40	6292.91	105.08	3.53

B. Carry Look Ahead Adder

This section proposes the implementation of 4-bit Carry Look Ahead Adder. A strategy for accelerating the addition process is by taking out inter stage carry delay. Such an addition operation is termed as look ahead-carry addition. The look ahead approach uses logic gates to look at he lower-order bits of augend and addend to check whether a higher order carry is to be generated. It makes use of two functions carry propagate (P_i) and carry generate (G_i) [10-12].

Where the equations for P_i and G_i can be expressed as:

$$P_i = A_i \oplus Bi \quad (2)$$

$$G_i = A_iB_i \quad (3)$$

For a 4-bit Carry look ahead adder (CLA) the carry output for each stage can be given as:

$$C_1=P_0C_0+G_0 \quad (4)$$

$$C_2=P_1C_1+G_1=P_1.(P_0C_0+G_0)+G_1 \quad (5)$$

$$C_3=P_2C_2+G_2=P_2.(P_1.(P_0C_0+G_0)+G_1)+G_2 \quad (6)$$

$$C_4=P_3C_3+G_3=P_3.(P_2.(P_1.(P_0C_0+G_0)+G_1)+G_2) +G_3 \quad (7)$$

The C_1 stage carry can be realized by using a m-GDI cell, by applying the P_0, the output of 1st partial full adder (PFA1) to the gate input of m-GDI cell, one input to the PFA i.e. C_0 as an input N of m-GDI cell and B_0 input of PFA as input P of m-GDI cell. The carry part of stage 1 i.e. C_1 is used for subsequent stage carry generation C_2, C_3 and C_4.

Fig. 21 stage shows the schematic of 1st partial full adder stage (PFA1) and Fig. 22 shows the schematic for C_1.

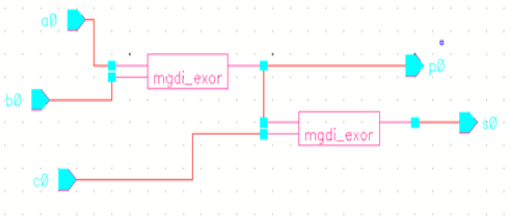


Fig. 21. 1st Partial Full Adder stage

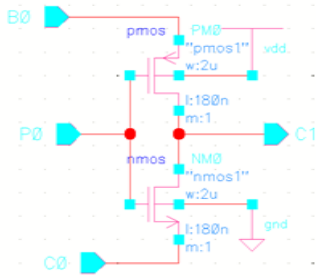
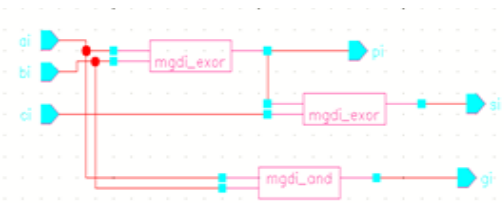


Fig. 22. Schematic of C_1 stage

To realize the sum for subsequent stages the partial full adder shown in the Fig. 23 is utilized.



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Fig. 23. Partial full adder for the S_1, S_2 & S_3 stage

Fig. 24 shows the schematic of m-GDI based 4-bit Carry Look Ahead Adder. Fig. 25 depicts the layout of 4-bit CMOS based CLA and Fig. 26 depicts the layout of m-GDI based 4-bit CLA. Table VI shows the performance analysis of 4-bit CLA.

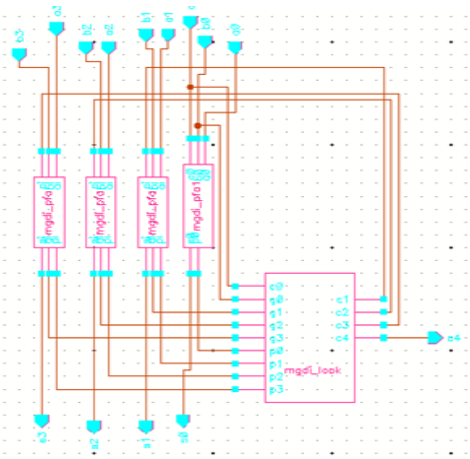


Fig. 24. Schematic of 4-bit m-GDI based CLA

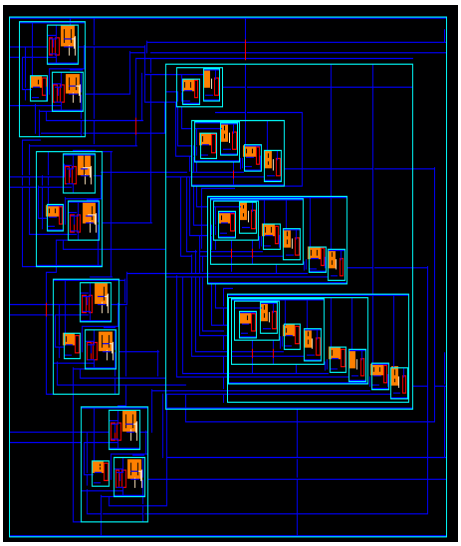


Fig. 25. Layout of CMOS based 4-bit CLA

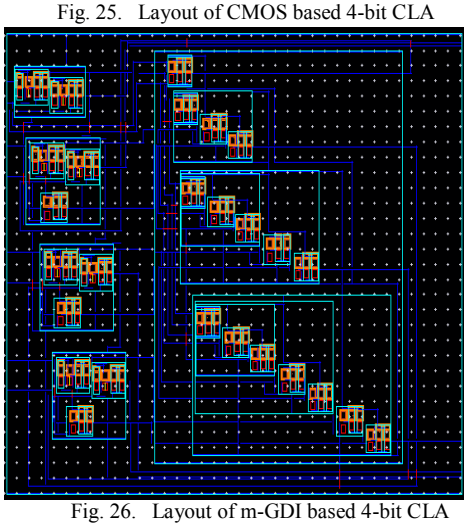


Fig. 26. Layout of m-GDI based 4-bit CLA

TABLE VI. PERFORMANCE ANALYSIS OF 4-BIT CLA

Design Approach	No. of Transistors	Area (μm^2)	Delay (psec)	Peak Power (μW)
CMOS	240	80350.7	261.60	6.38
m-GDI	70	27870.2	22.08	5.58

IV. DESIGN AND IMPLEMENTATION OF 8-BIT NOVEL ADDER ARCHITECTURE

The 8-bit hybrid adder is designed by cascading the two adder modules discussed in section III.

The lower 4-bits of the hybrid adder architecture uses 4-bit Carry Ripple Adder to carry out the summation of 4-LSB inputs to the adder. The carry output C_4 of the 4th stage of the CLA is applied as C_0 input to 1st stage of 4-bit Ripple Carry Adder. The 4-bit RCA calculates the sum of 4-MSB inputs to the adder. The final carry can be obtained at the carry output from the C_4 stage of the RCA.

Thus, the sum of lower 4-bits input $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are calculated employing the 4-bit CLA and the higher 4-bits input $A_7A_6A_5A_4$ and $B_7B_6B_5B_4$ are calculated using 4-bit RCA.

Fig. 27 illustrates the schematic of m-GDI based 8-bit Hybrid Adder Architecture and Fig. 28 depicts the layout of proposed 8-bit adder architecture using m-GDI technique.

Fig. 29 depicts the simulation waveforms of 8-bit proposed adder architecture for input combination '11111111' for $A_7A_6A_5A_4A_3A_2A_1A_0$ and '00000000' & '11111111' as input to $B_7B_6B_5B_4B_3B_2B_1B_0$.

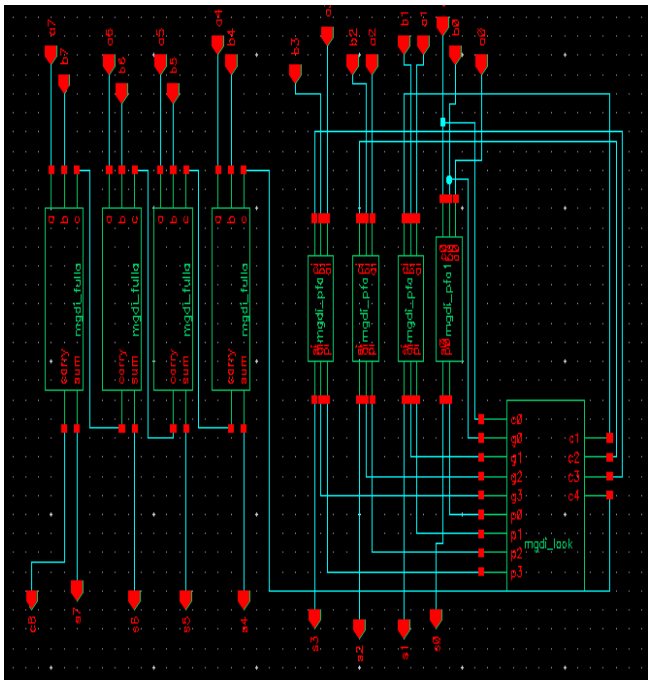


Fig. 27. Schematic of 8-bit Hybrid Adder architecture using m-GDI technique

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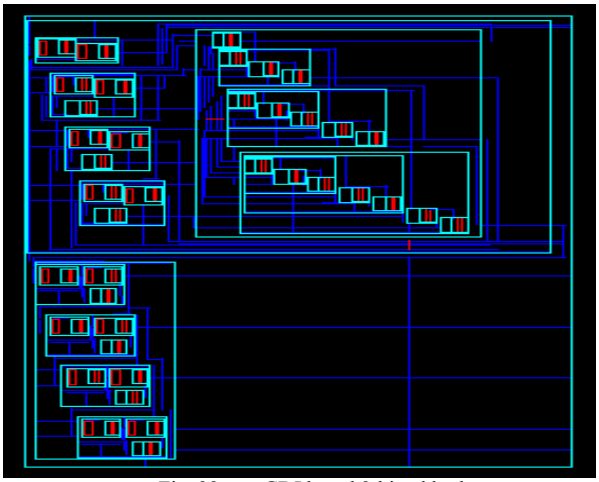


Fig. 28. m-GDI based 8-bit adder layout

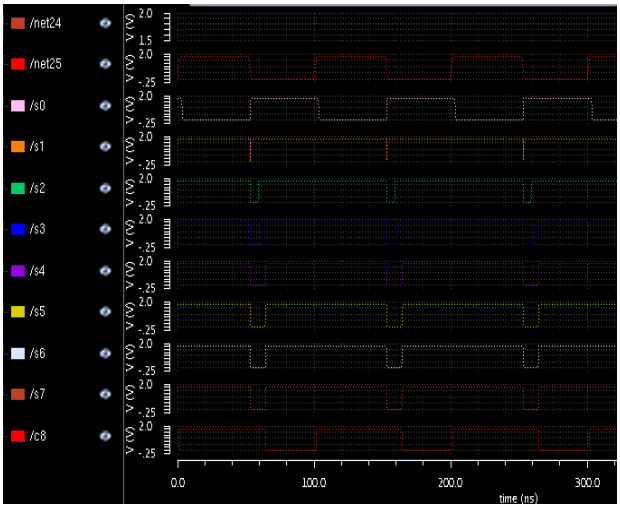


Fig. 29. m-GDI based 8-bit Adder simulation waveform

Table VII gives the performance analysis of 8-bit Hybrid Adder architecture.

TABLE VII. PERFORMANCE ANALYSIS OF 8-BIT NOVEL ADDER ARCHITECTURE.

Design Approach	No. of transistors	Area (μm^2)	Delay (psec)	Power (mW)
CMOS	408	191235.6	286.7	18.50
m-GDI	110	56638.3	82.58	12.05

V. CONCLUSION

This paper proposes a novel design of 4-bit Carry Look Ahead adder and 8-bit novel adder architecture. From the simulation result it was observed that the design of adder architectures using m-GDI technique results in high speed operation and consumes least no. of transistor. For ripple carry adder it was observed that the m-GDI based design consumes 81% lesser area, offers 72% lesser delay and the peak dynamic power reduced by 30% with respect to the traditional CMOS design.

Similarly, for the 4-bit carry look ahead adder it was observed that the m-GDI based design offered 91% lesser delay, a reduction in area by 65%, peak dynamic power reduced by 13% w.r.t the traditional CMOS design.

The proposed 8-bit hybrid adder architecture using m-GDI approach offered reduction in delay by 71%, the number of transistors reduced by a count of 298 when contrasted with the traditional CMOS design. The reduction in peak dynamic power was observed to be 35%. The reduction in area of the proposed adder architecture using m-GDI technique w.r.t the traditional CMOS implementation was found to be 70%.

Thus, from the above results it is clear that the design of 8-bit novel adder architecture using the mGDI technique results in lesser transistor count, dissipates lesser power and results in high speed operation.

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